Scalability & Future Plan

Date	9 July 2026
Team ID	XXXXXXXX
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, and extended features in the future. It also presents the roadmap for enhancing the Smart Agricultural Production Optimization Engine.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 1	Improve prediction accuracy using advanced ML	1-2 Months	Higher recommendation accuracy

Current System Limitations

S.No	Limitation	Impact	Priority
1	Limited dataset size	May reduce prediction accuracy for diverse regions	High
2	Single machine learning model	Limited flexibility in prediction performance	Medium
3	Local deployment only	Cannot support large-scale concurrent users	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited local users	Deploy on cloud with load balancing
2	Data Storage	CSV dataset	Use SQL/Cloud database for large datasets
3	Performance	Single model prediction	Optimize models and caching for faster response
4	Security	Basic Flask application	Add authentication, HTTPS and secure APIs

Future Roadmap

	models		
Phase 2	Cloud deployment with database integration	2-3 Months	Supports more users and scalability
Phase 3	Mobile application and multilingual support	3-5 Months	Better accessibility for farmers
Phase 4	IoT sensor integration for realtime monitoring	6+ Months	Smart farming and automated recommendations